

Predicting Readmission Rate for Diabetic Patients Using Machine Learning

Team Members and Emails

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Motivation

The hospital readmission rate is a major concern for both healthcare providers and patients, as it reflects the balance between reducing hospital costs and maintaining treatment efficiency and effectiveness. Among chronic conditions, diabetes is particularly challenging with complicated disease treatments and high readmission rates due to insufficient follow-up care and complex discharge procedures. Therefore, predicting which patients are likely to be readmitted soon after discharge has become a significant clinical focus and research priority. To address this, hospitals often target high-risk diabetes patients with interventions such as patient education programs, medication adjustments, or closer follow-up appointments to improve the quality of care and reduce preventable readmissions. The Centers for Medicare and Medicaid Services (CMS) launched the Hospital Readmissions Reduction Program (HRRP) on October 1, 2012. According to this program, hospitals that fail to control readmission rates face not only financial penalties but also public scrutiny. In this project, we aim to develop a predictive model that can identify patients at risk of complications before discharge. In addition to cost reduction and compliance with CMS regulations, our goal is to uncover which patient demographics, dosage changes, and other hospital-related features contribute the most to the likelihood of readmissions.

Data

The dataset used in this project originates from the University of California, Irvine (UCI) Machine Learning Repository (*Diabetes*), a well-known source for benchmark datasets in data science research. This dataset is widely studied in healthcare analytics as it provides valuable insights into risk factors and predictors of hospital readmission among diabetic patients. It contains 101,766 patient records collected between 1998 and 2008 across 130 different hospitals and integrated delivery networks in the United States. Each record corresponds to a patient diagnosed with diabetes who received laboratory tests, was prescribed medication, and experienced a hospital stay of up to 14 days. In total, the dataset provides 50 input features spanning both categorical and numerical variables. Our main focus will be on the binary target variable, *readmission*, which indicates whether a patient was readmitted or was not readmitted to the hospital.

Data Preprocessing

The raw dataset contained 50 features and 101,766 rows. One of the most significant changes during preprocessing was the transformation of categorical data into one-hot encoded data. For example, the original dataset has 20+ columns, each representing a medication, containing information about whether the drug was prescribed or if there was a change in the dosage. Values for these columns include *up*, *down*, *steady*, or *no*. These values needed to be transformed into up to four different dummy variables for each medication, resulting in an increase in the number of total columns. Additionally, there are 8 continuous variables focusing on prior number of hospital visits and medical history. These were all min-max normalized to handle different feature scales.

Some columns were dropped due to their irrelevance to readmission. For example, *encounter_id* and *patient_id* are unique identifiers that do not provide any value in the prediction of readmission. The *weight* column was dropped as more than 95% of the values were missing. The only other column that remains in the dataset with missing values is *race*. This is a categorical feature and imputation logic was applied to create a 'Missing' value. Lastly, there were also some duplicate patients who had multiple encounters (rows) in the dataset. The first instance of each patient was kept and the additional rows were dropped as we are focusing on first-time hospital visits. At the end of preprocessing, there are a total of 100 features and 71,518 samples containing categorical information on medication and demographics, and quantitative information on healthcare utilization and treatment intensity - such as number of procedures, time in hospital, etc. This leaves 60,159 training examples, 20,353 validation examples, and 20,353 testing examples, following a 60/20/20 data split.

Exploratory Data Analysis (EDA)

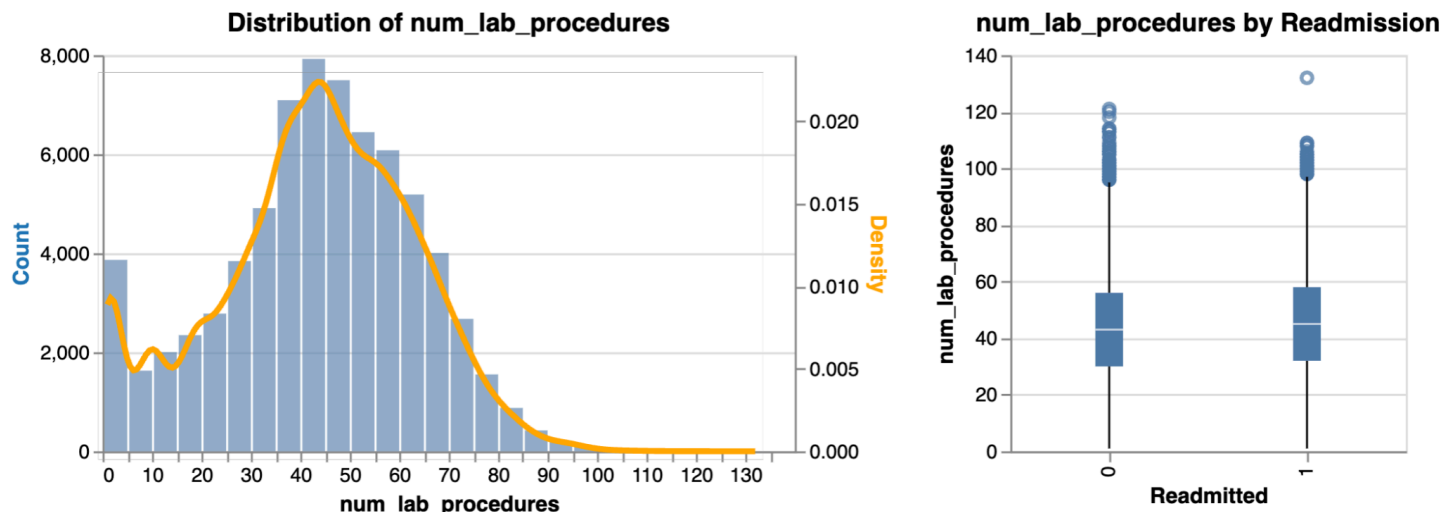
The exploratory data analysis provided a comprehensive understanding of the dataset's structure, feature distributions, and relationships with the target variable. The continuous variables, including *time_in_hospital*, *num_lab_procedures*, *num_procedures*, *num_medications*, and *number_diagnoses*, presented right-skewed distributions, indicating that most patients had fewer medical interventions and shorter hospitalizations; However, a small subset of the population underwent substantially higher utilization, which revealed chronic or severe cases. Health utilization indicators, such as *number_outpatient*, *number_emergency*, and *number_inpatient*, were heavily concentrated near zero, suggesting that the majority of patients had limited hospital encounters beyond their primary admission. The categorical features, including *race*, *gender*, and *age*, showcased demographic imbalances, with older and Caucasian patients overrepresented in our dataset, which may reflect underlying disparities in those who have access to healthcare. The binary treatment indicators, *change* and *diabetesMed*, revealed that starting or adjusting diabetes medication was linked to higher readmission rates.

Next, we performed correlation analysis using Pearson coefficients, which revealed positive associations among hospital utilization features, confirming that patients with more inpatient visits also tended to have longer stays. In addition, mutual information analysis identified *num_medications*, *number_inpatient*, and *time_in_hospital* as the strongest predictors of readmission, highlighting how

treatment complexity and intensity of care play a key role in patient outcomes. Overall, these findings suggest that frequent hospital use, medication adjustments, and complex clinical profiles are key risk factors for readmission.

Visualization

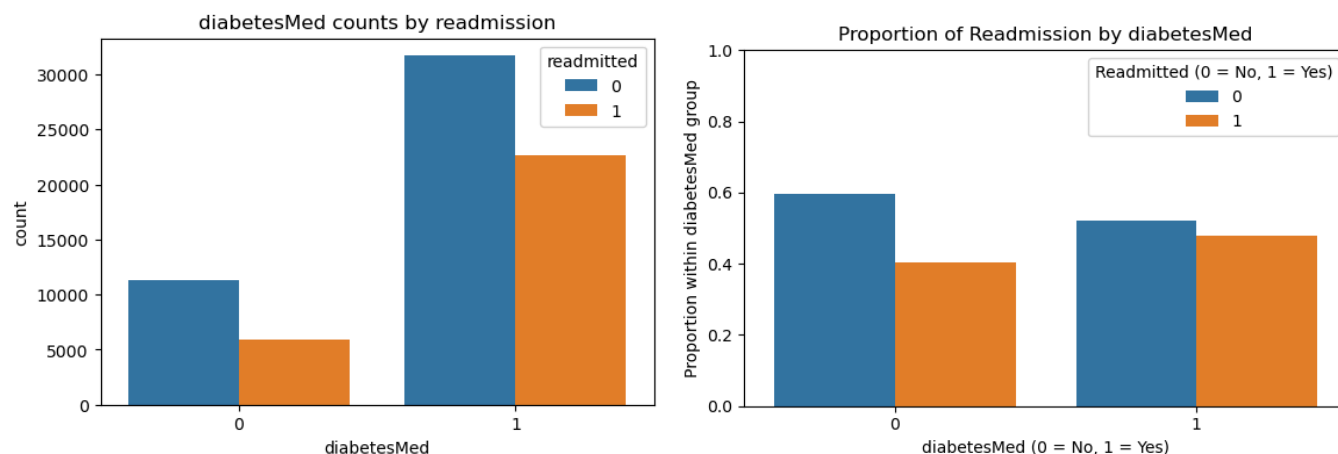
1. Distribution of each quantitative variable with Kernel Density overlay and Box plot of relation to Outcome.



Plotting the distribution of all of our quantitative features of interest showed a few concerning distributions. *num_inpatient*, *num_outpatient*, and *num_emergencies* showed heavy right (positive) skewness; all the other numerical features fell within the acceptable normal distribution range of -2 to +2. *Num_medications* is the closest to a mesokurtic distribution. The three positively skewed distributions referenced above are heavily leptokurtic, whilst the remaining variables show a platykurtic distribution.

Insight: Recommend either removing *num_outpatient*, *num_emergency*, and *num_inpatient* from the model, or taking a more aggressive standardization approach (perhaps log transformation).

2. Distribution of each of our categorical variables and their relation to our outcome variable



Insight:

All of our categorical variables showed a similar distribution across the target outcome classes, such that each categorical variable is similarly following the baseline distribution of our target feature. This points to each of our categorical features being weak individual predictors; however, this does not mean that when combined with each other or continuous features, they may show a strong predictive power.

The *diabetesMed* feature, however, is of interest as it indicates that in our patient population, **most patients discharged**—whether they were readmitted or not—had a diabetes medication prescribed during their hospital stay. Although the change group is the majority for all classes (so it doesn't separate the classes well, similar to the other categorical variables), the readmission rate for patients on diabetes medication (*diabetesMed=1*) is higher (48%) than the readmission rate for patients not on diabetes medication (*diabetesMed=0*), which is 40%. $P(\text{Readmitted}=1 \mid \text{diabetesMed}=1) \approx 0.48$

$P(\text{Readmitted}=1 \mid \text{diabetesMed}=0) \approx 0.40$. All else being equal, the variable diabetesMed is a good predictor of readmission because the conditional probability of the outcome ($\text{Readmitted}=1$) is not equal across the two groups.

Data Challenges

Our dataset contains multiple records per patient due to multiple visits. These duplicates can bias our model by overrepresenting some of the patients and distorting our feature importance. Additionally, some of our hospital utilization features were highly skewed due to extreme values, which could disproportionately influence our model predictions and must be addressed through appropriate transformations.

Methodology

This project aims to predict hospital readmission status for diabetic patients using Multiclass Logistic Regression (the baseline model), Random Forest, and XGBoost. The outcome variable has three possible labels—0 (“no readmission”), 1 (“readmitted over 30 days”), and 2 (“readmitted within 30 days”)—and will be one-hot encoded to support TensorFlow operations. These algorithms make sense for our problem because Multiclass Logistic Regression provides an interpretable linear baseline, while Random Forest and XGBoost capture nonlinear relationships and complex feature interactions in healthcare data.

Our baseline model will apply the top three parameters selected from the Mutual Information Scores in the EDA process. We will use TensorFlow with a softmax activation, categorical cross-entropy loss, and accuracy as the evaluation metric to create a multiclass classification model. The hyperparameters of this model—learning rate, batch size, and number of epochs—will be tuned and determined using the Keras Tuner function.

Random Forest and XGBoost will serve as our improved models after implementing the baseline model. We will utilize RandomizedSearchCV to tune hyperparameters such as the number of trees, maximum tree depth, and learning rate. After creating these two models, we will assess the top five important features and compare them with the features selected from the Mutual Information Scores plot.

Validation will involve computing the accuracy between the training and validation datasets. After model construction, we will evaluate each model’s performance by computing its corresponding accuracy value on the testing dataset. We will obtain the accuracy values from the assigned TensorFlow metric for the baseline model, and by calling the score and accuracy_score functions for Random Forest and XGBoost. The performance of the two improved models will be evaluated by comparing their accuracy values with that of the baseline TensorFlow model.

To compare and determine the best model, we plan to calculate the F1 scores ($2 * \text{precision} * \text{recall} / (\text{precision} + \text{recall})$) for each model after implementation. In addition, we will pay attention to the false positive rates for class 0 (“no readmission”), as failing to identify patients at risk of readmission could not only impose ethical and financial burdens on hospitals but also pose critical health risks to diabetic patients.

Contribution

- Colin Frishberg: Created Visualizations, helped with EDA and Data Processing
- Terra Jiang: Developed and implemented Multiclass Logistic Regression, Random Forest, and XGBoost models, and completed the Methodology section.
- Sai Sriya Mudigonda: Conducted exploratory data analysis, created visualizations for feature distributions, categorical trends, and correlations, and addressed the data challenges.
- Rahil Sharma: Completed the data overview and data preprocessing sections - feature selection, feature engineering, shuffling, train/val/test split.

Project GitHub Repo

https://github.com/UC-Berkeley-I-School/w207_final_project_frishberg_jiang_mudigonda_sharma